

Lecture 1: Cloud Computing – Overview

Computing can be defined as a means to solve any goal-oriented activity, including hardware, software systems, communications, and information gathering for a wide range of purposes. This definition encompasses a broad spectrum of resources and users, ranging from high-end researchers to the general public.

This lecture series focuses on **Cloud Computing** and its major components, recent trends, and future research opportunities. The initial lectures will provide an overview of cloud computing, covering topics like NIST models, characteristics, advantages, disadvantages, and the role of open standards.

Subsequent lectures will explore:

- **Cloud Computing Architecture**, including the typical computing stack and **Service-Oriented Architecture (SOA)**.
- **Service Models** available in the cloud, such as:
 - **Infrastructure as a Service (IaaS)**
 - **Platform as a Service (PaaS)**
 - **Software as a Service (SaaS)**
- **Deployment Models** of the cloud.
- **Service Management** in the cloud, including the concept of **Service Level Agreements (SLAs)** between the consumer and the **Cloud Service Provider (CSP)**. SLAs address aspects like service guarantees, resource requirements, availability, downtime, and **Quality of Service (QoS)**.
- **Cloudonomics**, which examines the economic model of cloud computing and factors such as resource costs, availability, and the economics of purchasing versus hiring resources.
- **Resource Management** from the CSP's perspective, focusing on managing resources like CPU, RAM, hard disk, and network connectivity to avoid overloading or underutilization.
- **Data Management**, including data storage, management, scalability, and data services in the cloud. This involves examining conventional data management approaches and their suitability for cloud environments, as well as the management of data stores and large-scale data processing. The lecture will also touch upon the architecture and challenges of data management, using examples like Dropbox.
- **Security** in the cloud, covering various aspects such as:
 - Infrastructure security
 - Data and storage security

- **Identity and Access Management (IAM)**
- Access control
- Trust, reputation, and risk management.
- Case studies and demos of open-source and commercial cloud platforms and simulators.
- **Recent Trends in Cloud Computing**, including concepts like **fog computing**.

The course structure will prioritize key concepts based on their importance. The lectures will build upon concepts from previous computing paradigms, including:

- **Distributed Computing**
- **Grid Computing**
- **Cluster Computing**
- **Utility Computing**.

Understanding the evolution of these computing paradigms will help contextualize the emergence of cloud computing.

Lecture 2: Cloud Computing – Overview (Contd.)

This lecture continues the discussion on the evolution of cloud computing.

Why Distributed Systems are Needed

- **Nature of the Application:** Some applications are computationally intensive, data-intensive, or require a high degree of robustness, making distributed systems suitable.
- **Performance:** Distributed systems offer enhanced performance for specific tasks by distributing the workload across multiple nodes.
- **Robustness:** They eliminate single points of failure by enabling tasks running on a failed node to be executed by other nodes, ensuring continuous operation.

Distributed Applications

- Distributed applications involve sets of processes spread across a network of machines that collaborate to solve a common problem.
- These applications are crucial for large-scale projects, where different operations are handled by different entities, ultimately achieving a unified objective.

- Examples of distributed applications include client-server systems and peer-to-peer computing, which facilitate resource management and truly distributed applications.

Grid Computing

- **Grid Computing** leverages the unused resources of networked computers to solve problems that are too demanding for standalone systems.
- It enables the virtualization of distributed computing and data resources, like processing power, bandwidth, and storage capacity, to create a single system image for seamless access.
- This approach offers dependable, consistent, and inexpensive access to vast computing capabilities.

Analogy to the Electrical Grid

- Similar to accessing power from an electrical grid, users or client applications can seamlessly access computing resources (processors, storage, data, applications) on a grid without needing to know their location.
- Grids connect various computing resources (PCs, workstations, etc.) and provide the mechanisms for accessing them.

Characteristics of Grid Computing

- Grids offer more than just information sharing; they focus on **sharing data and computing resources**.
- They facilitate efficient resource utilization across organizations and institutes by enabling the use of underutilized resources.
- Grids support the formation of local communities that share resources for specific purposes (e.g., biological sciences, genetic research).
- The interaction with underlying layers must be transparent and seamless for users.

Need for Grid Computing

- Grid computing finds significant applications in:
 - Scientific research, particularly data analysis, visualization, and collaboration
 - Computer simulation and modeling
- The increasing complexity of scientific and engineering problems demands more accurate and precise solutions, which grids can facilitate.
- Data visualization and the need to exploit underutilized resources further drive the adoption of grid computing.

Users of Grids

- Grid computing benefits various domains, including:
 - Physics
 - Weather forecasting
 - Material science
 - Reactor applications

Types of Grids

- **Computational Grid:** Focuses on computing power.
- **Data Grid:** Designed for data storage.
- **Collaborative Grid:** Supports collaborative research activities.
- **Network Grid:** Provides high-performance, fault-tolerant communication services.
- **Utility Grid:** Shares a wide range of resources, including data, computation cycles, and software.

Players in Grid Computing

- **Grid:** The central entity that provides resources and services.
- **Users:** Individuals or groups utilizing grid resources.
- **Sites:** Locations hosting heterogeneous resources.

The operation of grids involves policies for resource sharing and access control.

Cluster Computing

- **Cluster Computing** is a parallel or distributed computing platform where interconnected standalone computers work as a single integrated computing resource.
- Key components include:
 - Standalone computers (PCs, workstations, or SMPs)
 - Operating system
 - High-performance interconnects
 - Middleware
 - Parallel programming environment
 - Applications

Benefits of Clusters

- Clusters enhance speed and reliability compared to single computers.
- They typically have:
 - Faster and closer connections than a typical LAN
 - Low latency communication protocols
 - Looser coupling than SMPs

Types of Clusters

- **High-Availability or Failover Clusters:** Ensure high availability by transferring tasks to other nodes upon node failure.
- **Load-Balancing Clusters:** Distribute workloads across nodes for efficient resource utilization.
- **Parallel Distributed Processing Clusters:** Facilitate parallel and distributed processing tasks.

Utility Computing

- **Utility Computing** is a service provisioning model where a provider makes computing resources and infrastructure management available to customers on demand, charging for specific usage rather than a flat rate.
- This model, often referred to as "pay per use" or "pay as you go", is analogous to utilities like electricity or mobile phone services.

Utility Computing in Analogy to Electricity and Telecom Services

- Similar to metered electricity services, users pay for computing resources based on their consumption.
- Analogous to mobile phone services, users pay for services like calling, messaging, and data based on usage.

Key Features of Utility Computing

- **Metered Services:** Consumption of resources is tracked, and users are charged accordingly.
- **Pay-as-You-Go or Pay-as-You-Use Model:** Pricing is based on actual resource usage.
- **Data Center Virtualization and Provisioning:** Resources are virtualized and provided to users on demand, enabling customization and efficient utilization.
- **Outsourcing:** Services like software maintenance and data storage can be outsourced.
- **Shift from Data-Driven to Service-Driven Architecture:** Focus is placed on services rather than solely on data.

- **Automation:** Workflows can be automated, improving efficiency.

Advantages of Utility Computing

- **Lower Computing Costs**
- **Improved Performance**
- **Reduced Software Costs**
- **Instant Software Updates**
- **Improved Document Format Compatibility**
- **Unlimited Storage**
- **Increased Data Reliability**
- **Universal Information Access**
- **Access to the Latest Systems and Software**
- **Easier Group Collaboration**
- **Device Independence**

Risks and Disadvantages of Utility Computing

- **Data Backup Concerns:** Potential data loss in case of service provider failure.
- **Data Security Issues:** Risk of unauthorized data access.
- **Competence of Service Provider:** Uncertainty regarding the service provider's capabilities.
- **Challenges in Defining SLAs:** Difficulties in establishing mutually agreeable service level agreements.
- **Value from Chargebacks:** Concerns about the value obtained from chargeback mechanisms.

Cloud Computing

- **Cloud Computing** is a model that provides ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources.
- These resources, including networks, servers, storage, and applications, can be rapidly provisioned and released with minimal management effort or interaction with the service provider.

Cloud computing has evolved from earlier computing paradigms and aims to offer computing as a service, with features like pricing models, SLAs, security, and management, to meet diverse computing needs.

Lecture 3: Cloud Computing – Introduction

This lecture focuses on the introduction to cloud computing.

NIST Definition of Cloud Computing

- **Cloud Computing** is a model enabling convenient, on-demand network access to a shared pool of configurable computing resources (networks, servers, storage, and applications) that can be rapidly provisioned and released with minimal management effort.

Essential Characteristics of Cloud Computing

- **On-Demand Self-Service:** Users can provision resources independently without requiring human interaction with the service provider.
- **Broad Network Access:** Access to resources is available over a network, enabling ubiquitous access.
- **Resource Pooling:** Resources are pooled to serve multiple consumers in a multi-tenant model, with resources dynamically assigned and reassigned based on demand.
- **Measured Service:** Resource usage is monitored, controlled, and reported, providing transparency for both the provider and consumer.
- **Rapid Elasticity:** Resources can be rapidly and elastically provisioned and released to scale up or down based on demand.

Common Characteristics of Cloud Computing Platforms

- **Massive Scale:** Ability to scale resources up or down significantly.
- **Resilient Computing:** Ability to withstand and recover from failures.
- **Homogeneity:** Providing a consistent and uniform computing environment despite underlying heterogeneity.
- **Geographic Distribution:** Resources may be geographically dispersed across multiple locations.
- **Virtualization:** Abstraction of physical resources to create virtual instances.
- **Service Orientation:** Focus on services that communicate with each other, enabling a flexible and interoperable environment.
- **Low-Cost Software:** Reduced software costs due to the multi-tenant nature of cloud environments.
- **Advanced Security:** Enhanced security features related to access control, data preservation, and resilience against attacks.

Service-Oriented Architecture (SOA) in Cloud Computing

- **SOA** is a key aspect of cloud computing, where services are the primary means of communication and interaction.
- It involves:
 - **Service Provider:** Offers services.
 - **Service Consumer:** Utilizes services.
 - **Registry or Catalogue:** A central repository of available services.

Cloud Service Models

- **Software as a Service (SaaS):** Provides access to provider's applications running on cloud infrastructure. Users can access applications over the internet without managing the underlying infrastructure.
- **Infrastructure as a Service (IaaS):** Provides access to fundamental computing resources like servers, storage, and networks on demand. Users can configure and manage these resources as virtual instances.
- **Platform as a Service (PaaS):** Offers a platform for developing, testing, and deploying applications in the cloud. Users can focus on application development without managing the underlying infrastructure.

Cloud Deployment Models

- **Private Cloud:** Cloud infrastructure operated solely for a single organization.
- **Public Cloud:** Cloud services offered publicly over the internet by third-party providers.
- **Hybrid Cloud:** A combination of private and public cloud environments.
- **Community Cloud:** Shared infrastructure for a specific community of users with common interests or goals.

Virtualization and the Cloud

- **Virtualization** plays a crucial role in cloud computing.
- **Virtual Machine (VM)** technology enables running multiple VMs on a single physical machine, abstracting hardware resources and providing isolated environments.

Virtualization Stack

- **Bare Metal Hardware:** Physical hardware.
- **Virtual Machine Monitor (VMM) or Hypervisor:** Software layer that manages and emulates hardware resources for VMs.
- **Guest OS:** Operating system running inside a VM.

Advantages of Virtualization

- Running operating systems where physical hardware is unavailable.
- Simplified creation of new machines and backups.
- Facilitating software testing.

Cloud Sourcing

- **Cloud Sourcing** is driven by:
 - High-scale, low-cost service providers
 - Anytime, anywhere access via web browsers
 - Rapid scalability, incremental costs, and load sharing

Concerns and Disadvantages of Cloud Computing

- **Performance, Reliability, and SLAs:** Potential concerns about performance, reliability, and adherence to service level agreements.
- **Data Security:** Security and privacy concerns related to data stored in the cloud.
- **Service Parameters:** Ensuring and auditing service parameters like availability and data integrity.
- **Application Features and Choices:** Limited customization options compared to on-premise solutions.
- **Interoperability between Cloud Providers:** Lack of standardization and challenges in interoperability between different cloud platforms.
- **Standardization of APIs:** The need for standardized APIs for seamless communication and integration.
- **Security, Privacy, Compliance, Trust, Competence, and Risk:** Concerns regarding security, privacy, regulatory compliance, trust in providers, their competence, and potential risks.

Cloud Storage

- **Cloud Storage** is a crucial aspect of cloud computing, allowing data to be stored remotely and accessed over the internet.
- It offers advantages like lower computing costs, improved performance, reduced software costs, instant software updates, improved document compatibility, unlimited storage, increased reliability, universal access, and easier collaboration.
- However, it also presents disadvantages like the need for a constant internet connection, limited features, potential slowdowns, data loss risks, and dependence on the cloud provider's infrastructure.

Lecture 4: Cloud Computing Architecture

This lecture begins the discussion of cloud computing architecture.

Architectural Approach

- A high-level architectural approach to cloud computing involves aligning business goals with quality attributes and architectural tactics.
- **Business Goals** may include:
 - **Total Cost of Ownership (TCO)**
 - Stakeholder satisfaction
 - Compliance with standards
 - Market share
 - Flexibility
- **Quality Attributes** that support these goals include:
 - Availability
 - Elasticity
 - Interoperability
 - Security
 - Adoptability
 - Performance
 - Usability
 - Maintainability
- **Architectural Tactics** are employed to achieve the desired quality attributes, such as:
 - **Stateless Design**
 - **Loose Coupling**
 - **Heterogeneity**
 - **Interconnect and Homogeneity**
 - **Broad Network Access**
 - Caching mechanisms
 - Authentication (e.g., **Claim-Based Authentication**)

Building Blocks of Cloud Computing Architecture

- **Technical Architecture:**
 - Structuring according to the XaaS stack
 - Adopting cloud computing paradigms
 - Structuring cloud services and components
 - Middleware and communication
 - Management and security
- **Deployment and Operation Architecture:**
 - **Geolocation Checks**
 - Deployment issues
 - Operation and monitoring

XaaS Stack and Architectural Components

- The typical XaaS stack consists of IaaS at the bottom, followed by PaaS and SaaS.
- The architectural components include:
 - Storage and infrastructure
 - Applications
 - Client infrastructure
 - Middleware or platform
 - Cloud runtime libraries

Consumer Versus Provider View

- Different stakeholders have varying perspectives on the cloud architecture:
 - IT architects and operations focus on IaaS.
 - End-users and application administrators prioritize SaaS, such as CRM applications.
 - Software architects and developers work with PaaS, utilizing the infrastructure to develop and deploy applications.

Mapping to Popular Cloud Services

- Cloud services like Microsoft Windows Azure and Amazon EC2 align with the three-tier XaaS stack.
- Both platforms offer:
 - Infrastructure services at the bottom layer

- Middleware for platform services
- Applications at the top layer

Management and Security

- Management and security are crucial aspects of cloud architecture, spanning all layers of the XaaS stack vertically.
- The level of control and responsibility for management and security varies based on the service model:
 - IaaS: Responsibility primarily lies with the consumer after infrastructure provisioning.
 - SaaS: Responsibility primarily rests with the provider up to the application level.

Elasticity

- **Elasticity** refers to the ability to scale resources up or down on demand.
- Approaches to elasticity:
 - **Vertical Scale Up**: Adding more resources to a single computational unit.
 - **Horizontal Scale Out**: Distributing workload across multiple computational units.

The choice between vertical and horizontal scaling depends on cost, scalability, and fault tolerance requirements.

XaaS and Service-Oriented Architecture (SOA)

- **XaaS** is realized through the combination of **Service-Oriented Infrastructure (SOI)** and cloud computing principles.
- **Anything as a Service (XaaS)** is a generalization of cloud-related services, encompassing a growing number of services delivered over the internet.

XaaS Instances

- Common XaaS instances include:
 - **IaaS**: Raw infrastructure (CPU, memory, storage, etc.)
 - **PaaS**: Virtualized application development and runtime platform
 - **SaaS**: Software applications hosted in the cloud, available on a subscription basis
 - **Business Process as a Service (BPaaS)**: Horizontal or vertical business processes delivered on a subscription basis

Business Process as a Service (BPaaS)

- BPaaS involves providing entire business workflows as services, orchestrating multiple processes or applications to achieve a specific business objective.
- It facilitates the integration of sub-processes or sub-applications, potentially from different providers, to create a comprehensive solution.

Other XaaS Instances

- **Storage as a Service**
- **Security as a Service**
- **Database as a Service**
- **Monitoring and Management as a Service**
- **Communication and Content Computing as a Service**
- **Identity as a Service**
- **Backup as a Service**
- **Desktop as a Service**

Cloud Service Provider (CSP) Goals

- CSPs aim to:
 - Increase productivity
 - Enhance end-user satisfaction
 - Deliver innovative services
 - Maintain market position
 - Offer a flexible and configurable infrastructure

Classical IT Stack Versus Simplified IT Infrastructure

- The classical IT stack involves tightly coupled end-user devices, applications, and dedicated infrastructure.
- Cloud computing promotes a simplified IT infrastructure with greater flexibility, separating applications and infrastructure.
- This simplification allows for better resource utilization and potentially higher return on investment.

Client-Server Architecture Versus Service-Oriented Architecture (SOA)

- Traditional client-server architecture involves dedicated server programs that serve client requests.

- **SOA** offers a more flexible and decoupled approach, where services communicate with each other, facilitating interoperability and scalability.

Classical Cloud Stack

- The classical cloud stack logically consists of:
 - **Networking**: The foundation for communication and interaction.
 - **Storage**: A crucial component for data storage and management.
 - **Servers**: Physical or virtual machines that provide computing power.
 - **Virtualization**: Enables the creation of virtual instances of resources like servers, networks, and storage.
 - **Operating Systems**: Provide the environment for running applications.
 - **Runtime Libraries and Data**: Support application execution.
 - **Applications**: The top layer, providing the user-facing functionality.

The importance of each component varies based on the XaaS model being implemented.

Lecture 5: Cloud Computing Architecture (Contd.)

This lecture continues the discussion on cloud computing architecture, particularly focusing on service models, networking, and virtualization.

Client-Server Model and the Need for Cloud Computing

- The traditional client-server model involves a server that listens for requests from multiple clients.
- Cloud computing offers more complex and diverse service models, such as IaaS, PaaS, and SaaS, catering to a broader range of needs.
- While client-server architectures can be load balanced and scaled to some extent, cloud computing provides theoretically infinite scalability through virtualization.

Cloud Computing Service Model

- Cloud services are built upon a layered architecture:
 - **Infrastructure** (Compute, Network, Storage)
 - **Services** (IaaS, PaaS, SaaS)
 - **Business Processes**
 - **Presentation Layer** (User Interfaces, APIs)
- An **Integration Platform** connects these layers, facilitating:

- Service integration
- Orchestration
- **Quality of Service (QoS)** management ensures the desired performance and reliability.
- **Dissemination of Information** informs users about available services and how to access them.
- **Governance** encompasses management, legal, and policy aspects to ensure compliance and security.

Software as a Service (SaaS)

- SaaS provides web access to commercially available software, managed from a centralized location and delivered in a one-to-many model.
- It eliminates the need for users to handle upgrades and maintenance.
- **Application Programming Interfaces (APIs)** enable integration between different software components.

Use Cases for SaaS

- Applications with significant interactions between organizations and the outside world (e.g., email, marketing software)
- Applications requiring web and mobile access
- Applications needed for short-term or temporary use
- Applications experiencing significant demand spikes (e.g., tax filing, billing)

Scenarios Where SaaS May Not Be Ideal

- Applications demanding extremely fast processing or real-time data handling
- Applications with regulatory restrictions prohibiting external data hosting
- Applications where existing on-premise solutions meet all organizational needs

Platform as a Service (PaaS)

- PaaS delivers software development platforms over the web, enabling the creation and deployment of applications without managing the underlying infrastructure.

Characteristics of PaaS

- Integrated development environment (IDE) for development, deployment, hosting, and maintenance